

Spectral singularities in a non-Hermitian Friedrichs-Fano-Anderson model

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Spectral singularities are predicted to occur in a non-Hermitian extension of the Friedrichs-Fano-Anderson model describing the decay of a discrete state $|a\rangle$ coupled to a continuum of modes. A physical realization of the model, based on electronic or photonic transport in a semi-infinite tight-binding lattice with an imaginary impurity site at the lattice boundary, is proposed. The occurrence of the spectral singularities is shown to correspond either to a diverging reflection probability (for an amplifying impurity) or to a vanishing reflection probability (for an absorbing impurity) from the lattice boundary. In the former case, the spectral singularity of the resolvent is also responsible for the non-decay of state $|a\rangle$ into the continuum, in spite of the absence of bound states.

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I. INTRODUCTION.

Over the last decade, a great attention has been devoted to investigate the properties of non-Hermitian physical systems. In particular, it has been shown that the framework of quantum mechanics can be extended by relaxing the common constraint of Hermiticity for the underlying Hamiltonian H (see, for instance, [1–3] and references therein), provided that H has a real energy spectrum and is diagonalizable. In this case, after a proper change of the inner product (metric) of the Hilbert space, the non-Hermitian Hamiltonian H may be used to define a unitary quantum system [3]. Examples of such systems include non-Hermitian Hamiltonians with parity-time ($\mathcal{PT}$) symmetry [4], in which the reality of the energy spectrum (bound as well as radiation states) below a symmetry-breaking transition has been proved for several complex potentials [1]. Unfortunately, non-Hermitian Hamiltonians possessing a real-valued energy spectrum may fail to be diagonalizable because of the occurrence of exceptional points in the point spectrum [5, 6], or of spectral singularities in the continuous part of the energy spectrum [7–9]. Exceptional points refer to the coalescence of two or more discrete eigenvalues together with their eigenvectors; their physical relevance has been investigated in several works, and different physical realizations have been proposed and experimentally demonstrated [10]. On the other hand, spectral singularities refer to divergences of the resolvent operator $G(z) = (z - H)^{-1}$ belonging to the continuous spectrum of H , i.e. which do not correspond to square-integrable eigenfunctions. As opposed to exceptional points, spectral singularities have received less attention from physicists and have been mainly viewed as a curious mathematical property of certain non-Hermitian operators [7–9]. The physical meaning and relevance of spectral singularities have been highlighted solely quite recently, notably by A. Mostafazadeh [11] (see also [8, 9, 12, 13]) in the framework of wave scattering by complex potentials [14]. Mostafazadeh showed that spectral singularities of a non-Hermitian Hamiltonian with a complex potential

correspond to divergences of reflection and transmission coefficients of scattered states, i.e. to resonances with vanishing spectral width. He also investigated in details the appearance of spectral singularities in an electromagnetic realization of a non-Hermitian Hamiltonian with $\mathcal{PT}$ symmetry based on a waveguide filled by an atomic gas [11, 15], following an earlier proposal by Ruschhaupt and coworkers [16].

It is the aim of this work to investigate the onset of spectral singularities in a non-Hermitian extension of the famous Friedrichs-Fano-Anderson (FFA) model [17–19], which generally describes the decay of a discrete state coupled to a continuum of modes. The FFA model is encountered in different areas of physics, ranging from atomic physics [20–22] to quantum electrodynamics [23, 24] and condensed-matter physics [25–29]. The FFA model has been studied in the quantum theory of non-integrable systems [30] and used to describe unstable quantum systems, quantum mechanical decay and quantum Zeno dynamics [31, 32]. Simple models of single-particle electronic or photonic transport in tight-binding lattices can be also described by means of FFA Hamiltonians [25–29, 33–38]. In this work it is shown in particular that spectral singularities can be observed in photonic or electronic transport in semi-infinite tight-binding lattices with an imaginary impurity site at the lattice boundary, leading to either a diverging or a vanishing wave reflection from the lattice boundary.

The paper is organized as follows. In Sec.II we introduce a non-Hermitian extension of the FFA model, and derive the conditions for the appearance of spectral singularities. In particular, it is shown by direct calculations that the appearance of divergences of the resolvent operator $G(z)$ on the branch cut correspond to the non-diagonalizability of the FFA Hamiltonian. In Sec.III we present an example of non-Hermitian FFA model showing spectral singularities, which provides a simple model of electron or photonic transport in a semi-infinite tight-binding lattice with a boundary impurity site. It is shown that spectral singularities in this model correspond to either a vanishing or a diverging reflection probability from the lattice boundary. Finally, in Sec.IV the main conclu-